

Sam Harper - Curriculum Vitae

1 Personal Information

Current position

- Full Professor, Department of Epidemiology, Biostatistics & Occupational Health, McGill University

Education:

- PhD (*Epidemiologic Science*), University of Michigan Ann Arbor, USA 2005.
- MScPH (*Epidemiology*), University of South Carolina, USA 1999
- BA (*Biology*), Westminster College, USA, 1995

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2 Personal statement

My background is well suited to direct this project. I have a strong track record of research funding and productivity. Since 2019 I have led or co-led 5 research grants (both small and large) and generated substantial research output, with nearly 6000 citations since 2020. I have supervised and mentored over 15 trainees while also serving as Director of the PhD Program in Epidemiology at McGill from 2014-2024.

This proposal brings together two areas of research that have been passions of mine over the course of my career: equity and research transparency. Throughout my research career I have maintained a strong research interest in both the measurement and analysis of social inequalities in health, and I am eager to extend this work to assessing inequalities in how the grant peer system may (or may not) contribute to differences in funding success for vulnerable populations. I have also long maintained strong interests in metascience, reproducible research, and health equity, as noted below. Moreover, my interest in this project also derives from my personal experience. Over the past several years my service to CIHR in the capacity of Scientific Officer for the Public, Community and Population Health Committee has generated important questions for me regarding fairness in the peer review process for grants that I hope to explore in more detail with this proposal.

My work past integrating surveillance and policy on health inequalities has had an impact on policy and practice in Canada and internationally. I was a co-founding member of the McGill Public Policy and Population Health Observatory ([3PO](#)) since 2016. My international reputation as a scholar also led to an Endowed Professorship in *Impact of Health and Social Policy on Health Inequalities* at Erasmus University Medical Center, Rotterdam from 2018-2021.

3 Most significant contributions

Measuring and monitoring inequalities are crucial activities for providing evidence on the impact of interventions and policies on vulnerable populations, including for how structures such as grant peer review may disproportionately affect vulnerable subgroups. My prior work has provided key methodological insights and tools to advance the science of measuring health inequalities. Inequality is a value-laden construct, and my early work demonstrated the need to integrate ethical parameters into inequality measurement since different inequality metrics can provide diverging evidence on inequalities. This work led to two monographs solicited by the US National Cancer Institute and several highly-cited papers in top public health journals. My subsequent work in this area has been extended globally by my trainees.

My research on this topic has led to international recognition. The Pan American Health Organization (PAHO) translated one of my key papers into Spanish and my online health inequalities training [module](#)

into Spanish and Portuguese. I was chosen to serve as a member of the World Health Organization's (WHO) 2010-11 *Scientific Resource Group on Equity Analysis and Research* from over 450 nominations from around the world. I drafted the first WHO guidelines on monitoring social inequalities in health, which served as the foundation for the first WHO Handbook on Measuring Health Inequalities (Hosseinpoor *et al.*, 2013). I serve on the Steering Committee for the Public Health Agency of Canada's *Pan-Canadian Health Inequalities Reporting Initiative*. In 2022 I was invited to join the WHO's Expert Review Group for the development of the second edition of their *Health Inequality Monitoring Handbook*. I have been invited to provide technical advice and research workshops to PAHO, the US CDC, the Office of Environmental Justice at the US EPA, and the US National Cancer Institute. Most recently, I was invited by the US National Academy of Sciences, Engineering & Medicine to discuss my research on measuring inequalities for at their 2023 roundtable [series](#) on Population Health Improvement: *The Ecosystem of Health Equity Measures: A Workshop*.

Relevant recent papers:

- Chen D, Momen NC, Ejlskov L, Bødkergaard K, Werenberg Dreier J, et al. Socioeconomic inequalities in mortality associated with mental disorders: a population-based cohort study. *World Psychiatry* 2025;24(1):92-102.
- Hetherington E*, Darling E, Harper S, Nguyen F, Schummers L, et al. Inequalities in access to prenatal care during the COVID-19 pandemic: Analysis of a population-based cohort. *Paediatr Perinat Epidemiol* 2024;38:291–301.
- Carabali M, Harper S, Lima Neto AS, Dos Santos De Sousa G, Caprara A, Restrepo BN, et al. Decomposition of socioeconomic inequalities in arboviral diseases in Brazil and Colombia (2007-2017). *Trans R Soc Trop Med Hyg* 2022;116(8):717-726.
- Capurro DA*, Harper S. Socioeconomic inequalities in health care utilization in Paraguay: Description of trends from 1999 to 2018. *J Health Serv Res Policy* 2022;27(3):180-189.
- Sreeramareddy CT, Harper S. Trends in educational and wealth inequalities in adult tobacco use in Nepal 2001-2016: secondary data analyses of four Demographic and Health Surveys. *BMJ Open*. 2019 Sep 6;9(9):e029712.

Foundational papers:

- Harper S, Lynch J. Measuring and Decomposing Health Inequalities. In: Oakes JM, Kaufman JS (eds). *Methods in Social Epidemiology*, 2nd Edition; San Francisco, CA: Jossey-Bass, 2017, pp. 91-131.
- Harper S, King N, Meersman SC, Breen N, Reichman ME, Lynch J. Implicit value judgments in the measurement of health inequalities. *Milbank Quarterly* 2010;88:4-29.
- Harper S, Lynch J, Meersman SC, Breen N, Davis WW, Reichman ME. An overview of methods for monitoring social disparities in cancer with an example using trends in lung cancer incidence by socioeconomic position and race-ethnicity, 1992-2004. *Am J Epidemiol* 2008;167(8):889-99.
- Hosseinpoor AR et al. Handbook on health inequality monitoring: with a special focus on low-and middle-income countries. World Health Organization, 2013

Metascience and Reproducible Research

I also maintain a strong commitment to methodological transparency and reproducible science (Austin *et al.*, 2016; Hamra, Goldstein and Harper, 2019; Harper, 2019a; Goldstein, Hamra and Harper, 2020). When possible, I have made the raw data and statistical code to reproduce the tables and figures in my team's published papers freely available by uploading them to public repositories such as [Dataverse](#) and Open Science Foundation [page](#). I have uploaded data and code for more than a dozen of my published

papers, which have been downloaded over 12,000 times. I was invited to give a [workshop](#), *Reproducible Research: Why and How*, by the Society for Epidemiological Research, which covered scientific integrity problems, design and analytic solutions, dissemination and examples of how to conduct reproducible research. I also sought out more explicit training and took the Research Transparency and Reproducibility Training (RT2) course given by the Berkeley Institute for Transparent Social Science in 2019. Finally, I was selected and served as a 2019-2020 Faculty Fellow at [Project TIER](#) that focuses on Teaching Integrity in Empirical Research.

Relevant papers:

- Austin N*, Harper S, Kaufman JS, Hamra G. Challenges in Reproducing Results from Publicly Available Data: An Example of Sexual Orientation and Cardiovascular Disease Risk. *J Epidemiol Community Health* 2016;70(8):807-12.
- Goldstein ND, Hamra GB, Harper S. Are Descriptions of Methods Alone Sufficient for Study Reproducibility? An Example From the Cardiovascular Literature. *Epidemiology*. 2020 Mar;31(2):184-188.
- Hamra GB, Goldstein ND, Harper S. Resource Sharing to Improve Research Quality. *Journal of the American Heart Association*. 2019;8(15):e012292.
- Harper S. A future for observational epidemiology: Clarity, credibility, transparency. *Am J Epidemiol* 2019;188:840-845.

Evaluating Interventions

This project will also provide potential insights into how changes in the peer review system (e.g., virtual vs. in-person panels) may have affected funding. I have a substantial track record, much of it with my co-applicant Arijit Nandi, in conducting studies to estimate the impact of interventions on overall outcomes and inequalities. Thematically, this work all focuses on moving beyond surveillance and associations to get credible causal estimates of social and economic intervention effects, largely using quasi-experimental and experimental methods. I have conducted extensive work on the impacts of US mandatory seat belt laws on seatbelt use (Harper *et al.*, 2014), traffic crash deaths (Harper and Strumpf, 2017; Harper, 2019b), and police discrimination (Riddell *et al.*, 2020). I have also provided strong evidence on the impacts of medical marijuana legalization (Harper, Strumpf and Kaufman, 2012; Windle *et al.*, 2022), tobacco taxes (Manivong, Harper and Strumpf, 2017), abortion restrictions (Austin and Harper, 2019; Hawkins *et al.*, 2023), clinical obstetric interventions (Hutcheon *et al.*, 2017; 2020; 2022), as well as studies looking at the impact of economic recessions on health (Harper *et al.*, 2015; Harper and Bruckner, 2017).

To expand this work in 2016 I co-founded the Public Policy and Population Health Observatory with Arijit Nandi ([3PO](#)) at McGill to promote an integrated research agenda, create and publish policy databases, and evaluate the impacts of interventions on health. We have worked closely with academic and non-academic partners to develop and answer research questions and disseminate findings, as well as training doctoral students from McGill and six other countries. A large part of this work has produced high-profile papers on the impacts of paid leave policies (Nandi *et al.*, 2016; 2018). In addition to analysis of secondary data, through 3PO I have also (with partners) developed and implemented two large-scale randomized evaluations, one on daycare in India (Nandi *et al.*, 2020) and one on a complex maternal and newborn health intervention in Tanzania (Hetherington *et al.*, 2023). Our evidence on affordable daycare impacts in India was recently [featured](#) by MIT's Poverty Action Lab.

My work on the use and value of experimental and quasi-experimental designs for public health has led to international recognition from journals through solicited commentaries on the future of epidemiology (Harper and Strumpf, 2012; Harper, 2019a). I was invited to give the keynote address at the 2019 Cana-

dian Society for Epidemiology & Biostatistics Annual Meeting, as well as three international talks on the value of natural experiments for public health at [UC Berkeley](#), [Drexel](#), and [Erasmus University](#).

Relevant Recent Papers:

- Nandi A, Agarwal P, Chandrashekar A, Maloney S, Richardson R, Thakur L, Harper S. Access to affordable daycare and women's mental health in Rajasthan, India: Evidence from a cluster-randomised social intervention. *J Glob Health* 2024;14:04063.
- Hetherington E*, Harper S, Davidson R, Festo C, Lampkin N, Mtenga S, Teixeira C, Vincent I, Nandi A. Impact evaluation of the TAMANI project to improve maternal and child health in Tanzania. *J Epidemiol Community Health* 2023;77(6):410-6.
- Nandi A, Charters TJ, Quamruzzaman A, Strumpf EC, Kaufman JS, Heymann J, Mukherji A, Harper S. Health care services use, stillbirth, and neonatal and infant survival following implementation of the Maternal Health Voucher Scheme in Bangladesh: A difference-in-differences analysis of Bangladesh Demographic and Health Survey data, 2000 to 2016. *PLoS Med* 2022;19(8).
- Lopes FV*, Bakx P, Harper S, Ravesteijn B, Van Ourti T. The effects of supported housing for individuals with mental disorders. *Health Econ* 2022;31(S2):115-133.
- Hawkins SS, Harper S, Baum CF, Kaufman JS. Associations between State-Level Changes in Reproductive Health Services and Indicators of Severe Maternal Morbidity. *JAMA Pediatr* 2023;177(1):93-95.
- Harper S. Would stronger seat belt laws reduce motor vehicle crash deaths? A semi-Bayesian analysis. *Epidemiology* 2019;30:380-7.
- Harper S, Palayew A. The annual cannabis holiday and fatal traffic crashes. *Inj Prev* 2019;25:433–437.

4 Other contributions to knowledge

I have a strong record of service and my expertise is routinely sought by my peers. I have conducted over 180 peer reviews across a wide range of disciplines including medicine, epidemiology, public health, biostatistics, economics, and general social science. I have served on numerous grant panels for Canada (including several years as a Scientific Officer for CIHR) and international funding agencies, in addition to routinely serving a session organizer and reviewer of abstracts for annual meetings at the leading conferences in my discipline. I have also been identified as a *Top Reviewer* by Web of Science platform and earned an award as a *Top reviewers in Social Sciences - September 2019*.

Ad hoc reviews

Epidemiology and Public Health

American Journal of Public Health; Annals of Epidemiology; Epidemiology; International Journal of Epidemiology; Journal of Epidemiology & Community Health; Social Science & Medicine; Scandinavian Journal of Public Health; BMC Public Health; Paediatrics & Child Health; International Journal for Equity in Health; Caries Research; Journal of Urban Health; Health & Place; American Journal of Epidemiology; International Journal of Obesity; Cancer Management and Research; Open Medicine; International Journal of Pediatric Obesity; Cancer Epidemiology, Biomarkers & Prevention; British Journal of Cancer; BMC Cancer; European Journal of Public Health; Cancer Causes & Control; PLOS ONE; Addiction; Population Health Metrics; Global Health Action; Traffic Injury Prevention; Cancer; Oxford University Press (book proposal); Journal of Drug Issues; Environmental Health; Nature Communications Medicine; Nature; Nature Medicine; The Lancet Regional Health – Southeast Asia

General Medicine

Annals of Internal Medicine; BMJ (top 10% of peer reviewers in 2007); The Lancet; PLOS Medicine; JAMA; New England Journal of Medicine; Canadian Medical Association Journal; JAMA Pediatrics; Canadian J Cardiology

Health Policy

Milbank Quarterly; Health Affairs; Health Policy; Health Services Research; Evaluation Review

Social Science

Demography; Population Studies; Social Compass; Social Science History; Sociological Focus; Journal of Health and Social Behavior; Population Research and Policy Review; Journal of Health and Social Behavior

Biostatistics

Journal of the Royal Statistical Society (A); Statistics in Medicine

Economics

B.E. Journal of Economics and Policy; Health Economics; Oxford Development Studies; Journal of Economic Issues; Applied Economics Letters; American Journal of Economics & Sociology

Grant Reviews

Panel Member

2018-2023	Scientific Officer, Canadian Institutes for Health Research (CIHR), Public, Community & Population Health Committee
2017-2018	Member, Canadian Institutes for Health Research (CIHR), Public, Community & Population Health Committee

Ad hoc reviews

2021	Robert Wood Johnson Foundation Interdisciplinary Research Leaders Competition
2019	CIHR Canada Research Chairs (Tier 1)
2019	Faculty of Medicine, Internal grant review (2)

Recent Invited Lectures, Talks, Presentations

In the past 6 years I have also been invited to give international lectures on the history and application of multilevel models in social epidemiology, as well as talks on the measurement of health inequalities, and quasi-experimental study designs.

2024	Keynote Address: <i>Multilevel Models in Social Epidemiology: Past, Present, and Future</i> . Danish Epidemiology Society Annual Meeting, Aarhus University, Aarhus, DK Nov 7, 2024
2024	Invited Workshop: <i>Difference-in-Differences for Social Epidemiology</i> . Danish Epidemiology Society Annual Meeting, Aarhus University, Aarhus, DK Nov 6, 2024
2024	<i>Despair and Disadvantage: Some Questions</i> Presentation given at the MORTAL workshop, Nuffield College, Oxford University, UK Jun 10-11, 2024
2023	<i>Measuring Health Equity: Beyond Description</i> . US National Academy of Sciences, Engineering, & Medicine, The Ecosystem of Health Equity Measures: A Workshop, Oakland, CA, USA June 21, 2023 (remote)
2023	<i>A Quarter Century of Multilevel Models in Social Epidemiology: A Short Story in Three Acts</i> . Society for Epidemiologic Research Annual Meeting, Portland, OR, USA June 14, 2023
2020	Invited Workshop: <i>Reproducible Research: Why and How</i> . Pre-conference Workshop for the Society for Epidemiologic Research Annual Meeting (virtual) October 30, 2020

My expertise on measuring health inequalities has also been sought by national and international organizations, where I serve as a member of several standing committees:

2024-present	Member, Expert Review Group on <i>Collaborative Health Equity Measurement</i> , Maternal and Child Health Bureau, US Health Resources and Services Administration
2023-present	Member, Expert Review Group, <i>Health Inequality Monitoring Handbook</i> , World Health Organization
2017-present	Member, Working Group, <i>Health Equity Toolkit Assessment</i> , World Health Organization
2022-present	Steering Group Member, <i>Pan-Canadian Health Inequalities Reporting Initiative</i> , Public Health Agency of Canada
2019-2021	Working Group Member, <i>Health Inequalities Reporting Initiative</i> , Canadian Institute for Health Information

5 Supervisory and mentoring activities

I have strong track record of supervision. Below I list the postdocs, doctoral students, and master's degree students I have supervised or co-supervised in the past 6 years.

Postdoctoral fellows

2020-2023	Erin Hetherington
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Doctoral students

2022-present	Siobhan Carroll (co-supervised with Jill Baumgartner)
2021-present	Wenlu Yuan (co-supervised with Jill Baumgartner)
2021-present	Peter Socha (co-supervised with Jennifer Hutcheon)
2021-present	Talia Sternbach (co-supervised with Jill Baumgartner)
2021-present	Imen Farhat (co-supervised with Dimitra Panagiotoglu)
2020-2024	Francisca Vargas Lopez (Erasmus MC)
2018-2023	Walid Al-Soneidar
2017-present	Diego Capurro Fernandez
2014-2019	Nichole Austin

Master's degree students

2021-2022	Melia Alcantara (co-supervised with Seungmi Yang)
2020-2021	Peter Socha (co-supervised with Jennifer Hutcheon)
2018-2020	Hiba El-Haj (co-supervised with Shelley Clarke)
2019-2021	Talia Sternbach (co-supervised with Jill Baumgartner)

6 Other relevant information

None.

Bibliography

Austin, N. and Harper, S. (2019) "Quantifying the impact of targeted regulation of abortion provider laws on US abortion rates: a multi-state assessment," *Contraception*, 100(5), pp. 374–379. Available at: <https://doi.org/10.1016/j.contraception.2019.06.003>

Austin, N. *et al.* (2016) "Challenges in reproducing results from publicly available data: an example of sexual orientation and cardiovascular disease risk," *J Epidemiol Community Health*, 70(8), pp. 807–812. Available at: <https://doi.org/10.1136/jech-2015-206698>

Goldstein, N. D., Hamra, G. B. and Harper, S. (2020) "Are Descriptions of Methods Alone Sufficient for Study Reproducibility? An Example From the Cardiovascular Literature," *Epidemiology*, 31(2), pp. 184–188. Available at: <https://doi.org/10.1097/EDE.0000000000001149>

Hamra, G. B., Goldstein, N. D. and Harper, S. (2019) "Resource Sharing to Improve Research Quality," *J Am Heart Assoc*, 8(15), p. e12292. Available at: <https://doi.org/10.1161/JAHA.119.012292>

Harper, S. (2019a) "A Future for Observational Epidemiology: Clarity, Credibility, Transparency," *Am J Epidemiol*, 188(5), pp. 840–845. Available at: <https://doi.org/10.1093/aje/kwy280>

Harper, S. (2019b) "Would Stronger Seat Belt Laws Reduce Motor Vehicle Crash Deaths?: A Semi-Bayesian Analysis," *Epidemiology*, 30(3), pp. 380–387. Available at: <https://doi.org/10.1097/EDE.0000000000000990>

- Harper, S. and Bruckner, T. A. (2017) "Did the Great Recession increase suicides in the USA? Evidence from an interrupted time-series analysis," *Ann Epidemiol*, 27(7), pp. 409–414. Available at: <https://doi.org/10.1016/j.annepidem.2017.05.017>
- Harper, S. and Strumpf, E. C. (2012) "Social epidemiology: questionable answers and answerable questions," *Epidemiology*, 23(6), pp. 795–798. Available at: <https://doi.org/10.1097/EDE.0b013e31826d078d>
- Harper, S. and Strumpf, E. C. (2017) "Primary Enforcement of Mandatory Seat Belt Laws and Motor Vehicle Crash Deaths," *Am J Prev Med*, 53(2), pp. 176–183. Available at: <https://doi.org/10.1016/j.amepre.2017.02.003>
- Harper, S. *et al.* (2015) "Economic downturns and suicide mortality in the USA, 1980-2010: observational study," *Int J Epidemiol*, 44(3), pp. 956–966. Available at: <https://doi.org/10.1093/ije/dyv009>
- Harper, S., Strumpf, E. C. and Kaufman, J. S. (2012) "Do medical marijuana laws increase marijuana use? Replication study and extension," *Ann Epidemiol*, 22(3), pp. 207–212. Available at: <https://doi.org/10.1016/j.annepidem.2011.12.002>
- Harper, S. *et al.* (2014) "The effect of mandatory seat belt laws on seat belt use by socioeconomic position," *Journal of Policy Analysis and Management*, 33(1), pp. 141–161. Available at: <https://doi.org/10.1002/pam.21735>
- Hawkins, S. S. *et al.* (2023) "Associations Between State-Level Changes in Reproductive Health Services and Indicators of Severe Maternal Morbidity," *JAMA Pediatr*, 177(1), pp. 93–95. Available at: <https://doi.org/10.1001/jamapediatrics.2022.4136>
- Hetherington, E. *et al.* (2023) "Impact evaluation of the TAMANI project to improve maternal and child health in Tanzania," *J Epidemiol Community Health*, 77(6), pp. 410–416. Available at: <https://doi.org/10.1136/jech-2022-219995>
- Hosseinpoor, A. R. *et al.* (2013) *Handbook on health inequality monitoring: with a special focus on low- and middle-income countries*. World Health Organization
- Hutcheon, J. A. *et al.* (2020) "Antenatal corticosteroid administration and early school age child development: A regression discontinuity study in British Columbia, Canada," *PLoS Med*, 17(12), p. e1003435. Available at: <https://doi.org/10.1371/journal.pmed.1003435>
- Hutcheon, J. A. *et al.* (2017) "Safety of labour and delivery following closures of obstetric services in small community hospitals," *CMAJ*, 189(11), p. E431–E436. Available at: <https://doi.org/10.1503/cmaj.160461>
- Hutcheon, J. A. *et al.* (2022) "Antenatal corticosteroid administration and attention-deficit/hyperactivity disorder in childhood: a regression discontinuity study," *CMAJ*, 194(7), p. E235–E241. Available at: <https://doi.org/10.1503/cmaj.211491>
- Manivong, P., Harper, S. and Strumpf, E. (2017) "The contribution of excise cigarette taxes on the decline in youth smoking in Canada during the time of the Federal Tobacco Control Strategy (2002-2012)," *Can J Public Health*, 108(2), p. e117–e123. Available at: <https://doi.org/10.17269/cjph.108.5705>
- Nandi, A. *et al.* (2020) "Access to affordable daycare and women's economic opportunities: Evidence from a cluster-randomised intervention in India," *Journal of Development Effectiveness*, 12(3), pp. 219–239. Available at: <https://doi.org/10.1080/19439342.2020.1773898>

Nandi, A. *et al.* (2016) “Increased Duration of Paid Maternity Leave Lowers Infant Mortality in Low- and Middle-Income Countries: A Quasi-Experimental Study,” *PLoS Med*, 13(3), p. e1001985. Available at: <https://doi.org/10.1371/journal.pmed.1001985>

Nandi, A. *et al.* (2018) “The Impact of Parental and Medical Leave Policies on Socioeconomic and Health Outcomes in OECD Countries: A Systematic Review of the Empirical Literature,” *Milbank Q*, 96(3), pp. 434–471. Available at: <https://doi.org/10.1111/1468-0009.12340>

Riddell, C. A. *et al.* (2020) “Using change in a seat belt law to study racially-biased policing in South Carolina,” *Prev Med*, 130, p. 105884–105885. Available at: <https://doi.org/10.1016/j.ypmed.2019.105884>

Windle, S. B. *et al.* (2022) “The Impact of Cannabis Decriminalization and Legalization on Road Safety Outcomes: A Systematic Review,” *Am J Prev Med*, 63(6), pp. 1037–1052. Available at: <https://doi.org/10.1016/j.amepre.2022.07.012>